



Project title

by

Your Name

A dissertation submitted in fulfilment of the requirements of the degree of

Master of Commerce

in

Information Technology Management

at the

College of Business and Economics

UNIVERSITY OF JOHANNESBURG

Supervisor: Dr. Siyabonga Mhlongo

2026

Abstract

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Keywords: Keyword one, Keyword two, Keyword 3.

Declaration

I, Your Name, certify that the dissertation submitted by me for the Master of Commerce in Information Technology Management degree at the University of Johannesburg is my independent work and has not been submitted by me for a degree at another university.

YOUR NAME

(No signature required)

Dedication

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Acknowledgements

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Contents

Abstract	i
Declaration	ii
Dedication	iii
Acknowledgements	iv
List of Acronyms	vii
List of Figures	viii
1 Introduction	1
1.1 Introduction	1
1.2 Background and context	1
1.3 Rationale of the study	4
1.4 Problem statement	4
1.5 Significance of the study	6
1.6 Overview of the theoretical/conceptual framework	6
1.7 Research aim and objectives	6
1.8 Research questions	6
1.9 Scope and delimitation	6
1.10 Methodology overview	6
1.11 Ethical considerations	6
1.12 Dissertation structure	6
1.13 Chapter summary	6

2 Literature Review	7
3 Research Methodology	8
4 Results	9
5 Discussion	10
6 Conclusion	11
References	12
A Ethical Clearance Certificate	15
B Appendix Chapter	18
B.1 Appendix section	18
C Another Appendix Chapter	19

List of Figures

Chapter 1

Introduction

1.1 Introduction

This chapter introduces the study and also provides the background for examining the role of IT leadership in achieving ROI from AI coding assistants. It outlines the research problem being addressed and discusses the significance of the topic. Furthermore the chapter also presents the research questions and objections of this study. The final section of this chapter will provide an overview of the research methodology and theoretical framework used to guide the study.

1.2 Background and context

Artificial intelligence (AI) is one of the structural factors that has a significant impact on productivity and competitive advantages (Gillespie et al., 2025). As of 2023, generative AI systems have developed from being an experiment to reaching the phase of implementation within organizational processes (Gillespie et al., 2025). Recent reports from around the world show that organizations allocate more money to AI since they expect clear monetary outcomes, cost savings, and improved performance of organizational processes (Gillespie et al., 2025). Research on information systems and management has transcended from the stage of adoption to the understanding of added value meanings with the emphasis on differentiation of organizational value created from investments in AI systems as a function of organizational governance and management performance, not technology excellence (Hossain et al., 2025).

In software engineering, the use of generative AI tools are among the most noticeable forms of

AI-enabled augmentation (Peng et al., 2023). Providing developers with functionalities such as code suggestions and entire code functions in real time based on context, documentation, and debugging (Peng et al., 2023). Practical controlled experiments provide conclusive evidence that AI-assisted developers complete software development tasks faster and report increased productivity of about 55.8% compared with control groups (Peng et al., 2023). Collaboration between humans and AI systems is a paradigm shift from the traditional concept of automation, in which AI enhances developers' capabilities rather than replacing them with machines (Peng et al., 2023). Initial performance improvements demonstrated in controlled promising experiments may not necessarily guarantee long-term organizational performance benefits. Empirical studies report considerable variation in the productivity gains from AI-assisted software development tasks across software development teams and organizations.

As it has been long noted, organization culture and management style have been generally believed to affect the implementation success of new technologies. Within the digital field, values and practices will drive the culture of experimentation, learning, and data use in decision making as well as a positive attitude to tech adoption. New findings reveal that digital culture explains the linkage between human AI skills and innovation outcomes, and that culture readiness mediates the performance impact of AI Investments (An et al., 2024). Similarly, this applied also to employees' engagement in AI, ethical AI management or leadership impacts toward strategic alignment. Even though the significance of digital culture and leadership is being gradually recognized by most researchers, current research findings on AI Adoption is more fragmented (An et al., 2024). A significant number of recent works investigated AI adoption under the macro-organizational view or focused on the impacts for personal performance only instead of integrating digital culture, management and ROI in the software industry (Peng et al., 2023).

Generative AI (GenAI) has been increasingly adopted by African companies to address resource constraints and operational inefficiencies. Similarly, there is evidence that developing nations have tried extensively with AI and have positive views about AI's capabilities to boost productivity at work (Gillespie et al., 2025). Yet, literature to date highlights difficulties around digital skills pipelines, governance maturity and leadership capability among others in adopting GenAI technologies (Hughes et al., 2026). In this context, organizational culture and leadership remain particularly critical. If these factors are not supported by a clear strategic vision, trust and a learning orientation, AI adoption is likely to be less about value creation and more about

segmentation or symbolism. The digital transformation landscape in Africa is a valid environment for exploring the impact of socio-organizational factors on the achievement of returns from AI technologies.

With regards to software engineering in South Africa, the environment provides a relevant framework in grasping how the various organizations adopt and utilize advanced AI tools. Several studies confirm the fact that AI has emerged as one of the major components that define the transformation of operations on companies and the way they develop their culture due to the increase in number of firms using AI technologies in their business processes (Murire, 2024). There are opportunities and challenges that needs to be addressed in order to leverage organizational performance through AI adoption and alignment with organizational culture and structure, as indicated by research reviews on AI tool adoption in organizations (Romeo & Lacko, 2025). Though the adoption of AI technology affects many aspects of the organization's performance positively, like productivity and output, the effect of AI technology on aspects like the organization's culture and employees morale and engagement is complicated and depends on many factors, as indicated by the study on the impact of AI tools in workplace settings (Kassa & Worku, 2025). Studies on the effect of generative AI tools on software engineering have indicated that AI technology positively affects organizations by enhancing productivity in coding and other software engineering activities through employees adoption of AI tools (Alenezi & Akour, 2025).

Leadership and organizational culture play an important role in the interpretation, adoption and usage of AI technologies by work teams. Research on AI adoption has found that the adoption of AI related technology is also dependent on leadership styles that facilitate embedding technology initiatives into organizational culture (Romeo & Lacko, 2025). Research has also found that the adoption of AI-related technology is dependent on the adoption of the organizational culture that is conducive to the adoption of technology (Kassa & Worku, 2025). Further more research on organizational culture has also shown that cultural elements, particularly the norms that define innovation, collaboration and adaptability, have an impact on the use of AI by staff (Murire, 2024). In the case of software engineering teams that use AI-based coding assistants for various functions, both leadership as well as culture determine the effectiveness of AI in terms of productivity or quality and financial returns.

1.3 Rationale of the study

coming soon

1.4 Problem statement

Generative AI was primarily used for research and information retrieval purposes in its initial years; however, the use of generative AI has shifted towards the process of knowledge work itself. Experimental results show that generative AI enhances the efficiency and effectiveness of cognitive work (Noy & Zhang, 2023). Furthermore, studies conducted in a work environment setting have also shown improvements in productivity as well as employees performance due to the use of generative AI. It seems from these results that the benefit of generative AI depends upon its implementation rather than the technology itself (Brynjolfsson et al., 2025). More experimental results regarding software development can be found in favour of this theory. Large-scale researches conducted on GitHub Copilot have been showing improvements in the completion time as well as in the efficiency of the development process of developers (Cui et al., 2026). Furthermore, controlled experiments using AI programming assistants have also showed improvements in the speed of completing software development tasks, this provides empirical support for the ROI of generative AI in coding assistance (Peng et al., 2023). All these results collectively show that AI coding assistants are capable of increasing productivity. It is therefore necessary to conduct more research in order to investigate the effects of organizational factors on this phenomenon.

While all these improvements are measurable, research evidence in existing literature has proved that the performance impact varies based on the situation. In software engineering, research has been able to demonstrate that the code generated using GitHub Copilot increases productivity but at the same time creates a level of inconsistency in accuracy, maintainability, and effort required for review purposes; and this clearly demonstrates the need to depend on the management and integration of the system (Moradi Dakhel et al., 2023). Research on the governance of artificial intelligence in organizations has suggested that to achieve the sustainability of value creation, there should be processes and systems in place to ensure accountability and supervision of the leadership to convert experiments into performance improvement (Mäntymäki et al., 2022). Further more studies on organizational innovation with generative AI has demonstrated that the organization will be able to achieve various performance impacts depending on whether

the AI is used for automation or augmentation purposes; and this demonstrates the importance of strategic leadership and culture (Holmström & Carroll, 2025). Overall, this existing research evidence shows the role of organizational culture and strategic leadership in determining whether the improvements in productivity due to AI coding assistants can be sustained and converted to measurable return on investment.

The importance of such mechanisms can also be seen within the context of Africa due to the fact that organizations face limitations in terms of skills development, governance, and digitization. The empirical study on the readiness of AI within Africa demonstrated the level of readiness and the organization's ability to implement AI which demonstrates the importance of organizations in the transition from AI pilot to its performance (Isagah & Ben Dhaou, 2024). In the context of South Africa, empirical studies on the financial sector have demonstrated the role of organizational factors, including leadership support and readiness in the adoption of new and emerging technologies which demonstrate the importance of such factors in achieving ROI from using AI coding assistants in software development teams (Matsepe & Van der Lingen, 2022). The empirical research conducted in South African environment highlights the significance of sociotechnical systems to AI adoption and change management as mechanisms of achieving performance outcomes related to AI adoption (Smit et al., 2024). Also the research reveals the significance of managerial perceptions towards AI adoption in achieving performance outcomes within the field of human resources management as the field of AI application.

Collectively, the global evidence shows that AI coding assistants can improve productivity, while organisational research suggests that sustained ROI is influenced by strategic alignment, effective governance, and organisational culture. The African and South African evidence base also indicates that the context is significant in relation to the successful adoption of technology. It is because of this that the evidence base offers a rationale for conducting an in-depth exploration in relation to the role of culture and leadership in achieving ROI with AI coding assistants.

1.5 Significance of the study

1.6 Overview of the theoretical/conceptual framework

1.7 Research aim and objectives

1.8 Research questions

1.9 Scope and delimitation

1.10 Methodology overview

1.11 Ethical considerations

1.12 Dissertation structure

1.13 Chapter summary

Chapter 2

Literature Review

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Chapter 3

Research Methodology

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Chapter 4

Results

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Chapter 5

Discussion

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Chapter 6

Conclusion

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Appendix A

Ethical Clearance Certificate

**CBE RESEARCH ETHICS COMMITTEE**

Dear Researchers

ETHICAL APPROVAL GRANTED FOR RESEARCH PROJECT

Decision: Clearance granted

This letter serves to confirm that the proposed research project indicated in the table below, has been reviewed by the Department of Applied Information Systems at the University of Johannesburg. Ethical clearance is hereby granted and is valid for three years, from 01 January 1900 until 31 December 2999.

Applicant	
Supervisor	
Student/staff number	
Title	
Decision date at meeting	
Reviewers	
Ethical clearance code	
Rating of application	

CODE 01 – Approved

CODE 02 – Approved with suggestions/requirements
with no re-submission

CODE 03 – Referred back

CODE 04 – Disapproved, cannot re-submit

The researcher/s may now commence with the study providing that:

1. The researcher/s will ensure that the project adheres to ethical research requirements
2. The researcher/s will be conducting the study as set out in the approval application
3. The researcher/s will ensure that project adheres to all applicable legislation, scopes of practice, professional codes of conduct and scientific standards as it pertains to the field or study.
4. The researcher/s will bring under the attention of the research ethics committee any proposed changes, concerns that arise, and unexpected ethical management issues
5. Any changes that can affect the study-related risks for participants or researchers must be reported to the committee in writing
6. Only data collected during the three valid period is permissible and valid for the study

7. No data collect before or after the three valid period is ethically valid and the Research Ethics Committee takes no responsibility for any issues arising in this situation
8. No fieldwork activities may continue after the ethical clearance has expired. A request for an extension of ethical clearance can be made in writing to the REC
9. All data collection must be done within the Government regulations of the lockdown
10. The researcher/s must be aware that COVID-19 may influence the responses and needs to be controlled for

Furthermore, the researcher/s must please address the following comments from the ethics committee:

- N/A

Prof M Name

AIS REC

aisrec@uj.ac.za

Appendix B

Appendix Chapter

B.1 Appendix section

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Appendix C

Another Appendix Chapter

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